

# Data Analytics with GenAI – Tableau Mini Project

## Key Findings :-

### 1. Overall Sales Performance & Town-wise Trends :-

- The analysis shows significant variation in real-estate sales across towns. The total selected sales metric shown in the Tableau view is approximately \$234.63 billion across the displayed towns.
- **Hot Springs** is the strongest-performing town in the Sales by Town chart, with approximately **\$9.63 billion** in sales. It is followed by towns such as Asheville, Morgantown, Oxford, Columbia, Bend, and Flagstaff, which also show relatively high sales amount.
- The sales trend chart also reveals substantial variation across months. In particular, there is a very large spike around August, reaching approximately \$5.5 billion for a town-level series.

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### 2. Town-wise Geographic Performance :-

- The Sales Map confirms that real-estate activity is distributed across multiple locations throughout the United States, with differences in the size of the sales markers indicating differences in sales volume.
- The largest markers represent locations with comparatively higher sales values, while smaller markers indicate lower sales volumes.
- The Sales by Town chart supports this geographic pattern, with Hot Springs reaching approximately \$9.63B, significantly higher than many of the other displayed towns.

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### 3. Sales Ratio Reveals Significant Property-Type Outliers :-

- The Sales Ratio Heat Map shows that most Town–Residential Type combinations have sales ratios approximately in the 0.7–1.3 range, indicating relatively consistent performance across many combinations.
- However, there are several notable outliers.

For example:

- Bellingham – Single Family: approximately 37.15
- Keene – Single Family: approximately 21.30
- Several other combinations, particularly involving the Null category, also show relatively high ratios.

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### 4. Single Family Properties Dominate the Property Composition :-

- The Property Type Distribution chart shows a strong concentration in a small number of property categories.
- The largest category is Single Family, with approximately 273,147 properties, representing roughly 55% of the clearly labeled property records shown in the chart.

Other major categories include:

- Residential: 112,099
- Condo: 68,789
- Apartments: 34,172

- Three Family: 7,889
- Apartments: 943
- Public Utility: 8

This indicates that the dataset is heavily dominated by Single Family properties, followed by Residential and Condo categories.

Key takeaway: Single Family properties are the dominant property segment and should therefore receive significant attention when analyzing overall real-estate market behavior.

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## **5. Recommendations for Business & Further Analysis :-**

Based on the combined findings, the analysis suggests several opportunities:

- Prioritize high-performing towns such as Hot Springs and other towns appearing near the top of the Sales by Town ranking.
- Focus on Single Family properties, as they represent the largest share of the property records.
- Investigate extreme Sales Ratio values, particularly Bellingham–Single Family and Keene–Single Family, before using them for strategic decisions.
- Examine the large monthly spike more closely at the Town level, because the trend chart contains multiple town-level series and the extreme August value may be driven by a specific town rather than the entire market.